Week 4 Notes

Mathematics I: Introduction to Polynomials

Course: Mathematics I — IIT Madras BS Programme

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Lecture 1 — What Is a Polynomial?

From Layman's Intuition to a Precise Mathematical Definition

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1 A Layman's Perspective

Informal Idea

A polynomial is a **sum of mathematical terms**, where each term can be:

- a **number** (e.g., 10),
- a **variable** (e.g., x),
- a **product of several variables** (e.g., x^2y).

Example: $x^2 + 4y^2 + 2z + 10$ — sum of a constant (10), and terms involving three variables. Looks like a polynomial.

Tempting but wrong example: $t^{1/2} + t$ — a Layman might accept this too ("it's a sum of two variable-like terms"). We will see why this **fails** under the precise definition.

2 The Mathematician's Precise Definition

Formal Definition of a Polynomial

A **polynomial** is an algebraic expression in which the only operations on the variables are **addition, subtraction, multiplication**, and the variables appear with **natural number (non-negative integer) exponents** only.

Natural exponents: $0, 1, 2, 3, \ldots$ (i.e., $\mathbb{N}_0$, sometimes called whole numbers or non-negative integers).

Crucially forbidden:

- Fractional/rational exponents: $x^{1/2} = \sqrt{x}$ — **not allowed**.
- Negative exponents: $x^{-1} = 1/x$ — **not allowed**.
- Variables in denominators (in simplest form), under radicals, or inside transcendental functions.

3 Why $t^{1/2} + t$ Fails

The Rational Exponent Problem

In $t^{1/2} + t$, the term $t^{1/2}$ has exponent $\frac{1}{2}$, which is a **rational number**, not a natural number.

4 Why the Name "Polynomial"?

Etymology

Poly (Greek) = **many**. **Nomen** (Latin) = **name** or **term**.
Polynomial = "many terms."

Number of terms	Name	Example
1	Monomial	$3x^2$
2	Binomial	$x + 4$
3	Trinomial	$x^2 + 4x + 2$
≥ 4 (or general)	Polynomial	$x^3 + 2x^2 - x + 5$

Convention in this course: We do **not** distinguish monomial/binomial/trinomial from "polynomial." A monomial is treated as a special case of a polynomial (a polynomial with one term).

5 Testing Expressions: Is It a Polynomial?

Worked Examples

Expression	Polynomial?	Reason
3	Yes	Constant term alone is a valid monomial
$3x^2$	Yes	Number $\times$ variable with natural exponent
$x^2 + 4y^2 + 2z + 10$	Yes	Sum of valid monomials, all natural exponents
$x^2 + 4x + 2$	Yes	Each term valid (product of variables, number$\times$variable, constant)
$x + x^{1/2}$	**No**	Second term has rational exponent $\frac{1}{2}$
$x + y + xy + x^3$	Yes	x, y, xy, x^3 all valid; check each term individually

Method: Check Term by Term

1. Break the expression into individual terms (monomials).
2. For each term, check: is it a number, a variable, or a product of variables with **natural number exponents**?
3. If **every** term passes, the expression is a polynomial.
4. If **any** term fails (rational/negative exponent, variable in denominator (after solving it in simplest form) , etc.), it is **not** a polynomial.

6 General Form: Polynomial in One Variable

Standard Notation

A polynomial in one variable x of degree n:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0$$

Using summation notation:

$$\boxed{P(x) = \sum_{m=0}^{n} a_m x^m}$$

Terminology:

- a_m: **coefficient** of the term x^m.
- m: the **exponent** (must be a natural number for a valid polynomial term).
- $a_n \neq 0$ ensures the polynomial has a well-defined highest term (otherwise it could "extend to infinity" ambiguously).

Example: For $P(x) = x^4+1$: rewriting as $\sum_{m=0}^{4} a_m x^m$, we have $a_4 = 1$, $a_0 = 1$, and $a_1 = a_2 = a_3 = 0$.

7 Restriction to Real Coefficients

In this course, we work with polynomials having **real coefficients** (i.e., each $a_m \in \mathbb{R}$).
Other branches of mathematics study polynomials with:

- **Integer coefficients** ($a_m \in \mathbb{Z}$),
- **Complex coefficients** ($a_m \in \mathbb{C}$).

These are outside the scope of this course.

8 Polynomials in Multiple Variables

Type	Example	Notes
One variable	$x^2 + 4x + 2$	$\sum_{m=0}^{n} a_m x^m$
Two variables	$x^2 + 4y^2 + 2$	Coefficients indexed by two exponents, e.g., a_{mn}
Three+ variables	$x + y + z + x^3$	Any number of variables, natural exponents throughout

9 Key Takeaways — Lecture 1

1. A **polynomial** is a sum of terms, each being a number, a variable, or a product of variables — with all exponents restricted to **natural numbers** $(0, 1, 2, 3, \ldots)$.
2. Rational exponents (e.g., $x^{1/2}$) or negative exponents (e.g., x^{-1}) **disqualify** an expression from being a polynomial.
3. **Etymology**: poly (many) + nomen (terms) = "many terms." Monomial (1 term), binomial (2), trinomial (3) are all treated as polynomials in this course.

4. **Test method**: check every term individually; if any term violates the natural-exponent rule, the whole expression fails.
5. **Standard form**: $P(x) = \sum_{m=0}^{n} a_m x^m$ with $a_n \neq 0$; a_m are coefficients.
6. This course restricts to polynomials with **real coefficients**. Polynomials can have one, two, or more variables.

Lecture 2 — Degree of a Polynomial

Degree of a Variable, Degree of a Term, and Classification by Degree

1 Motivating Example

Consider the polynomial:

$$P(x, y) = 3x^2 + 4x^2y^2 + 10y + 1$$

To understand its **degree**, we build up from the degree of a variable, to the degree of a term, to the degree of the whole polynomial.

2 Degree of a Variable in a Term

Definition

The **degree of a variable** in a term is the **exponent** on that variable within that term.
Example: In the term $4x^2y^2$:

- Degree of x in this term $= 2$.
- Degree of y in this term $= 2$.

3 Degree of a Term

Definition

The **degree of a term** (monomial) is the **sum** of the degrees of all variables appearing in that term.
Example: For the term $4x^2y^2$:

$$\text{degree} = \deg(x) + \deg(y) = 2 + 2 = 4$$

4 Degree of a Polynomial

Definition

The **degree of a polynomial** is the **largest** degree among all its terms with **nonzero coefficients**.

Working Through the Example

$P(x, y) = 3x^2 + 4x^2y^2 + 10y + 1$. Analyze each term:

Term	Rewritten	Degree calculation	Degree
$3x^2$	$3x^2y^0$	$\deg(x)+\deg(y)=2+0$	2
$4x^2y^2$	—	$\deg(x)+\deg(y)=2+2$	4
$10y$	$10x^0y^1$	$\deg(x)+\deg(y)=0+1$	1
1	$1\cdot x^0y^0$	$\deg(x)+\deg(y)=0+0$	0

Key trick: A pure x-term has an implicit y^0; a constant has both x^0 and y^0. This lets every term be compared on the same footing.

$$\text{Degrees found: } \{2,4,1,0\} \quad \Rightarrow \quad \text{largest} = 4$$

$$\boxed{\deg(P)=4}$$

5 The Zero Polynomial: An Exception

Degree of the Zero Polynomial Is Undefined

Consider the polynomial that is identically 0:

$$0 = 0 + 0\cdot x + 0\cdot x^2 + 0\cdot x^3 + \cdots$$

Every coefficient is zero. Since the degree requires a term with a **nonzero** coefficient, there is no valid "largest degree" to pick.

$$\boxed{\deg(Zero-Polynomial) \text{ is } \textbf{undefined}}$$

Why this matters: This subtlety becomes important later when working with the **division algorithm** for polynomials.

6 Classification of Polynomials by Degree

Standard Names by Degree

Degree	Name	Form (1 variable)	Condition
0	Constant	a_0	$a_0 \neq 0$
1	Linear	$a_1x + a_0$	$a_1 \neq 0$
2	Quadratic	$a_2x^2 + a_1x + a_0$	$a_2 \neq 0$
3	Cubic	$a_3x^3 + a_2x^2 + a_1x + a_0$	$a_3 \neq 0$
4	Quartic	$a_4x^4 + \cdots + a_0$	$a_4 \neq 0$
5	Quintic (Quantic)	$a_5x^5 + \cdots + a_0$	$a_5 \neq 0$
n (general)	Polynomial (degree n)	$\sum_{m=0}^{n} a_m x^m$	$a_n \neq 0$

Degree, Not Variable Count, Determines Classification

The names (linear, quadratic, cubic, ...) depend on the **degree**, regardless of how many variables are involved.

Examples:

- $x + y$: two variables, but degree 1 (each term has degree 1) $\Rightarrow$ still called **linear**.
- $x^2 + y^2$: two variables, degree 2 $\Rightarrow$ **quadratic**.
- A degree-3 polynomial in any number of variables is called **cubic**.

Using precise classification ("this is a quadratic polynomial") conveys more information than the generic term "polynomial."

7 Key Takeaways — Lecture 2

1. **Degree of a variable** in a term = its exponent in that term.
2. **Degree of a term** = sum of the degrees of all variables in that term.
3. **Degree of a polynomial** = largest degree among all terms with nonzero coefficients.
4. **Trick for multi-variable terms**: treat missing variables as raised to the power 0 to compare degrees uniformly.
5. **Zero polynomial**: degree is **undefined** (no nonzero-coefficient term exists). Important for the division algorithm later.
6. **Classification by degree**: constant (0), linear (1), quadratic (2), cubic (3), quartic (4), quintic (5), ... — classification depends on **degree**, not the number of variables.

Lecture 3 — Addition & Subtraction of Polynomials

Combining Like Terms and the General Algorithm

1 Polynomials as Functions

A degree-n polynomial in one variable:

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0$$

can be viewed as a **function** $P : \mathbb{R} \to \mathbb{R}$.

Range Depends on the Polynomial

- **Degree 1** ($P_1(x) = a_1 x + a_0$, $a_1 \neq 0$): a line. Range $= \mathbb{R}$ (entire real line).
- **Degree 1 with** $a_1 = 0$: a horizontal line (constant). Range = single value.
- **Degree 2** ($P_2(x) = a_2 x^2 + a_1 x + a_0$): a parabola. Range depends on sign of a_2 (opens up $\to$ minimum onward; opens down $\to$ maximum backward).

We focus here on the **algebra** of polynomials (addition, subtraction); graphing comes later.

2 Addition of Polynomials: Building Intuition

Example 1: Polynomial + Constant

$P(x) = x^2 + 4x + 4, \quad Q(x) = 10$

Rewrite $Q(x)$ with matching powers: $Q(x) = 0x^2 + 0x + 10$.

$$P(x) + Q(x) = (1+0)x^2 + (4+0)x + (4+10) = x^2 + 4x + 14$$

Degree check: $\deg(P) = 2$, $\deg(Q) = 0$. Since $2 > 0$, $\deg(P+Q) = 2 = \max(2, 0)$.

Example 2: No Clashing Terms

$P(x) = x^4 + x^3, \quad Q(x) = 4x + 1$

Rewrite both with all powers from 0 to 4:

$$P(x) = 1x^4 + 1x^3 + 0x^2 + 0x + 0, \qquad Q(x) = 0x^4 + 0x^3 + 0x^2 + 4x + 1$$

Add coefficient-wise:

$$P(x) + Q(x) = (1{+}0)x^4 + (1{+}0)x^3 + (0{+}0)x^2 + (0{+}4)x + (0{+}1) = x^4 + x^3 + 4x + 1$$

Degree check: $\deg(P) = 4$, $\deg(Q) = 1$. Result: $\deg(P+Q) = \max\{4, 1\}$.

Example 3: With Clashing (Overlapping) Terms

$P(x) = x^3 + 2x^2 + x, \quad Q(x) = x^2 + 2x + 2$

Both have x^2 and x terms — these **clash** and must be combined:

$$P(x) + Q(x) = x^3 + (2{+}1)x^2 + (1{+}2)x + (0{+}2) = x^3 + 3x^2 + 3x + 2$$

Degree check: $\deg(P) = 3$, $\deg(Q) = 2$. Result: $\deg(P + Q) = 3 = \max\{3, 2\}$.

3 The Degree Rule for Addition

Degree of a Sum

If $\deg(P) = m$ and $\deg(Q) = n$:

$$\boxed{\deg(P + Q) = \max\{m, n\}} \qquad (\text{when } m \neq n)$$

- If $m = n$: result **may** has degree n (because the leading terms could get canceled)

Caution: If $m = n$ and the leading coefficients happen to be negatives of each other (e.g., $a_n = -b_n$), the degree-n terms cancel, and the resultant degree could be **less than** n.

4 General Formula for Addition

Formal Definition

Let $P(x) = \sum_{k=0}^{n} a_k x^k$ (with $a_n \neq 0$) and $Q(x) = \sum_{j=0}^{m} b_j x^j$ (with $b_m \neq 0$).

Step 1: Let $N = \max\{m, n\}$.
Step 2: Pad the polynomial of lower degree with zero coefficients up to degree N.
Step 3: Add coefficient-wise:

$$\boxed{P(x) + Q(x) = \sum_{k=0}^{N} (a_k + b_k)\, x^k}$$

5 Subtraction of Polynomials

Subtraction = Addition of the Negative

$$P(x) - Q(x) = P(x) + \big(-Q(x)\big)$$

Multiply every coefficient of $Q(x)$ by -1, then add using the same algorithm as before.

$$\boxed{P(x) - Q(x) = \sum_{k=0}^{N} (a_k - b_k)\, x^k}$$

Revisiting the Three Examples with Subtraction

Polynomials	$P(x)+Q(x)$	$P(x)-Q(x)$
$x^2+4x+4, \quad 10$	$x^2+4x+14$	x^2+4x-6
$x^4+x^3, \quad 4x+1$	x^4+x^3+4x+1	x^4+x^3-4x-1
$x^3+2x^2+x, \quad x^2+2x+2$	x^3+3x^2+3x+2	x^3+x^2-x-2

Key observation: Since multiplying a nonzero constant polynomial by -1 doesn't change its degree, the **same degree rule** ($\deg = \max\{m,n\}$, generically) applies to subtraction as well.

6 Key Takeaways — Lecture 3

1. A degree-n polynomial can be viewed as a function $P:\mathbb{R}\to\mathbb{R}$; its range depends on the specific polynomial (line, parabola, etc.).
2. **Addition recipe**: pad both polynomials to the same highest degree with zero coefficients, then add coefficients of matching powers.
3. **Degree of sum**: $\deg(P+Q)=\max\{\deg P,\deg Q\}$ when the degrees differ. If degrees are equal, leading terms may cancel, potentially lowering the degree.
4. **General formula**: $P(x)+Q(x)=\sum_{k=0}^{N}(a_k+b_k)x^k$ where $N=\max\{m,n\}$.
5. **Subtraction** = addition after negating every coefficient of the second polynomial: $P(x)-Q(x)=\sum_{k=0}^{N}(a_k-b_k)x^k$.
6. Multiplying by -1 does not change degree (for nonzero polynomials), so the same degree rule applies to both addition and subtraction.
7. This algorithmic approach (pad, then add/subtract coefficient-wise) is directly suitable for computer implementation.

Lecture 4 — Multiplication of Polynomials

From FOIL to a General Coefficient Formula

1 Beyond FOIL

The **FOIL method** (First, Outer, Inner, Last) multiplies two **binomials**:

$$(ax+b)(cx+d) = acx^2 + adx + bcx + bd$$

But FOIL only works for binomial × binomial. We need a method for polynomials of **arbitrary degree**.

2 Step 1: Multiplying by a Monomial

Example: $P(x) = x^2 + x + 1, \quad Q(x) = 2x^3$

Multiply each term of $P(x)$ by the monomial $2x^3$, using the **law of exponents** ($x^a \cdot x^b = x^{a+b}$):

$$2x^3(x^2 + x + 1) = 2x^{2+3} + 2x^{1+3} + 2x^3 = 2x^5 + 2x^4 + 2x^3$$

3 Step 2: Extending FOIL — Binomial × Trinomial

Example: $(2x+1)(x^2+x+1)$

Split $(2x+1)$ into two monomials and multiply **each** into the trinomial separately:

$$2x(x^2 + x + 1) = 2x^3 + 2x^2 + 2x$$

$$1(x^2 + x + 1) = x^2 + x + 1$$

Now add these two results by matching powers of x (same rule as polynomial addition):

$$2x^3 + (2{+}1)x^2 + (2{+}1)x + 1 = 2x^3 + 3x^2 + 3x + 1$$

Key idea: Multiplication of polynomials = **term-by-term multiplication**, followed by **addition** of the resulting polynomials (matching powers).

4 Finding the Pattern: Coefficient of x^k

Consider $P(x) = a_2x^2 + a_1x + a_0$ and $Q(x) = b_1x + b_0$ (none of the coefficients zero). Multiplying term by term:

$$P(x)Q(x) = a_2b_1x^3 + a_1b_1x^2 + a_0b_1x + a_2b_0x^2 + a_1b_0x + a_0b_0$$

Spotting the Pattern

Group by power of x and look at the **subscripts**:

- Coefficient of x^3: only a_2b_1 — subscripts $2+1=3$. ✓
- Coefficient of x^2: $a_1b_1 + a_2b_0$ — subscripts $1{+}1=2$ and $2{+}0=2$. ✓
- Coefficient of x^1: $a_0b_1 + a_1b_0$ — subscripts $0{+}1=1$ and $1{+}0=1$. ✓
- Coefficient of x^0: only a_0b_0 — subscripts $0+0=0$. ✓

Rule discovered: The coefficient of x^k collects **all pairs** a_jb_{k-j} where the subscripts sum to k.

5 The General Multiplication Formula

Product of Two Polynomials

Let $P(x) = \sum_{i=0}^{n} a_i x^i$ (degree n) and $Q(x) = \sum_{i=0}^{m} b_i x^i$ (degree m).

Then:

$$\boxed{P(x) \cdot Q(x) = \sum_{k=0}^{m+n} c_k x^k, \qquad \text{where } c_k = \sum_{j=0}^{k} a_j\, b_{k-j}}$$

Degree of the product: $\deg(P \cdot Q) = m + n$ (**always**, provided $a_n \neq 0$ and $b_m \neq 0$; unlike addition, degrees always add, they don't just take the max).

Convention: any coefficient not explicitly listed (i.e., beyond the polynomial's degree) is taken to be 0.

6 Worked Example: Two Quadratics

$P(x) = x^2 + x + 1, \quad Q(x) = x^2 + 2x + 1$

Coefficients: $a_0 = 1,\ a_1 = 1,\ a_2 = 1$ (and $a_3 = a_4 = \cdots = 0$); $\quad b_0 = 1,\ b_1 = 2,\ b_2 = 1$ (and $b_3 = b_4 = \cdots = 0$).

Since $\deg(P) = \deg(Q) = 2$, the product has degree $2 + 2 = 4$. Compute $c_k = \sum_{j=0}^{k} a_j b_{k-j}$ for $k = 0, \ldots, 4$:

k	**Sum** $c_k = \sum_j a_j b_{k-j}$	**Value**
0	a_0b_0	$1 \times 1 = 1$
1	$a_0b_1 + a_1b_0$	$1{\times}2 + 1{\times}1 = 3$
2	$a_0b_2 + a_1b_1 + a_2b_0$	$1{\times}1 + 1{\times}2 + 1{\times}1 = 4$
3	$a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0$	$0 + 1{\times}1 + 1{\times}2 + 0 = 3$
4	$a_0b_4 + a_1b_3 + a_2b_2 + a_3b_1 + a_4b_0$	$0 + 0 + 1{\times}1 + 0 + 0 = 1$

$$\boxed{P(x) \cdot Q(x) = x^4 + 3x^3 + 4x^2 + 3x + 1}$$

7 Two Equivalent Methods

Method	Description
Term-by-term	Multiply every term of $Q(x)$ into every term of $P(x)$; then add matching powers. Intuitive, good for small polynomials.
Coefficient formula	$c_k = \sum_{j=0}^{k} a_j b_{k-j}$. Systematic, ideal for programming/computation.

8 Key Facts About Multiplication

- The product of two polynomials is **always a polynomial** (closure property).
- $\deg(P \cdot Q) = \deg(P) + \deg(Q)$ — degrees **add** (contrast with addition, where degrees take the **max**).
- **Division** of two polynomials, unlike multiplication, does **not** always yield a polynomial (this will be explored in the next lecture).

9 Key Takeaways — Lecture 4

1. **FOIL** handles binomial $\times$ binomial only. For general polynomials, use **term-by-term multiplication** + addition of results.
2. Multiplying by a monomial uses the **law of exponents**: $x^a \cdot x^b = x^{a+b}$.
3. **Pattern**: the coefficient of x^k in a product is $c_k = \sum_{j=0}^{k} a_j b_{k-j}$ — all pairs of subscripts summing to k.
4. **General formula**: $P(x)Q(x) = \sum_{k=0}^{m+n} c_k x^k$ with c_k as above.
5. **Degree of product**: $\deg(P \cdot Q) = \deg(P) + \deg(Q)$ (degrees **add**, unlike addition where degrees take the max).
6. Multiplication of polynomials always yields a polynomial; division does **not** always (topic of the next lecture).

Lecture 5 — Division of Polynomials

Long Division, Quotient, and Remainder

1 A Necessary Condition for Division

Degree Requirement

To divide polynomial $P(x)$ (numerator, degree m) by $Q(x)$ (denominator, degree n), $Q(x) \neq 0$, for a non-trivial long division method to possible, we generally need:

$$\boxed{m \geq n}$$

Example of impossibility: Dividing the constant 4 (degree 0) by $2x + 1$ (degree 1) is **not possible** as a polynomial division — the numerator's degree is less than that of denominator. The expression must be left as a fraction: $\dfrac{4}{2x+1}$.

2 Case 1: Dividing by a Constant

$$\frac{a_2x^2 + a_1x + a_0}{c} = \frac{a_2}{c}x^2 + \frac{a_1}{c}x + \frac{a_0}{c}$$

Simply divide every coefficient by c. Always gives back a polynomial.

3 Case 2: Dividing by a Monomial

Example: $(3x^2 + 4x + 3) \div x$

Split into separate terms and divide each:

$$\frac{3x^2}{x} + \frac{4x}{x} + \frac{3}{x} = 3x + 4 + \frac{3}{x}$$

Key point: The term $\dfrac{3}{x}$ cannot be simplified further — the numerator (3, degree 0) has **smaller degree** than the denominator (x, degree 1). It remains as a fraction in the final answer.

Conclusion: Division of polynomials does **not always yield a polynomial** — unlike addition, subtraction, and multiplication.

4 Case 3: Dividing by a Binomial — Using Factorization

Example: $(3x^2 + 4x + 1) \div (x + 1)$

Step 1: Divide leading terms: $\dfrac{3x^2}{x} = 3x$.
Step 2: Multiply back: $3x(x + 1) = 3x^2 + 3x$.
Step 3: Split $4x$ as $3x + x$ to match:

$$3x^2 + 4x + 1 = \underbrace{3x^2 + 3x}_{=3x(x+1)} + x + 1 = 3x(x + 1) + 1(x + 1) = (3x + 1)(x + 1)$$

Result: $\dfrac{3x^2 + 4x + 1}{x + 1} = 3x + 1$ (exact division — $x + 1$ is a **factor**).

What If It's Not an Exact Factor?

Example: $(3x^2 + 4x + 4) \div (x + 1)$

Same process: $3x^2 + 4x + 4 = 3x(x + 1) + (x + 4) = 3x(x + 1) + 1(x + 1) + 3$

$$= (3x + 1)(x + 1) + 3$$

Result:

$$\frac{3x^2 + 4x + 4}{x + 1} = 3x + 1 + \frac{3}{x + 1}$$

Here $x + 1$ is **not** a factor, so a nonzero **remainder** (3) is left over.

5 The General Long Division Algorithm

Algorithm (Illustrated with a Bigger Example)

Divide $P(x) = x^4 + 2x^2 + 3x + 2$ by $Q(x) = x^2 + x + 1$.
Setup: Write $P(x)$ with **all** powers present (insert $0x^3$ where missing):

$$x^4 + 0x^3 + 2x^2 + 3x + 2$$

Repeat until the remainder's degree is less than $\deg(Q)$:

1. Divide the **leading term** of the current remainder by the leading term of $Q(x)$.
2. Multiply this result by the **entire** $Q(x)$.
3. **Subtract** from the current remainder.
4. The new remainder becomes the working expression; repeat.

Step-by-Step Execution

Round	Divide leading terms	Multiply back	New remainder
1	$x^4/x^2 = x^2$	$x^2(x^2+x+1) = x^4+x^3+x^2$	$-x^3 + x^2 + 3x + 2$
2	$-x^3/x^2 = -x$	$-x(x^2+x+1) = -x^3-x^2-x$	$2x^2 + 4x + 2$
3	$2x^2/x^2 = 2$	$2(x^2+x+1) = 2x^2+2x+2$	$2x$

Stop: The remainder $2x$ has degree $1 < \deg(Q) = 2$. Division ends here.

Final Answer

$$\boxed{\frac{x^4 + 2x^2 + 3x + 2}{x^2 + x + 1} = x^2 - x + 2 + \frac{2x}{x^2 + x + 1}}$$

Quotient: $x^2 - x + 2$. **Remainder**: $2x$.

6 The Division Statement

General Form of Polynomial Division

For polynomials $P(x)$ (dividend) and $Q(x) \neq 0$ (divisor), there exist unique polynomials $S(x)$ (quotient) and $R(x)$ (remainder) such that:

$$\boxed{P(x) = S(x) \cdot Q(x) + R(x)}, \qquad \deg(R) < \deg(Q) \text{ or } R(x) = 0$$

Equivalently: $\dfrac{P(x)}{Q(x)} = S(x) + \dfrac{R(x)}{Q(x)}$.

If $R(x) = 0$: $Q(x)$ is a **factor** of $P(x)$, and the division is **exact**.

7 Key Contrast: Division vs. Other Operations

Operation	Result always a polynomial?
Addition	Yes
Subtraction	Yes
Multiplication	Yes
Division	**No** — may leave a remainder term $R(x)/Q(x)$

8 Key Takeaways — Lecture 5

1. Polynomial division generally requires deg(numerator) $\geq$ deg(denominator).
2. Dividing by a **constant**: divide every coefficient. Dividing by a **monomial**: split into terms; some may not simplify (leftover fraction).

3. **Factorization approach**: split middle terms cleverly so the divisor factors out completely (when it's an exact factor).
4. **Long division algorithm**: repeatedly divide leading terms, multiply back, subtract, until remainder's degree $<$ divisor's degree.
5. **Division statement**: $P(x) = S(x)Q(x) + R(x)$ with $\deg(R) < \deg(Q)$. This is the polynomial analogue of integer division with remainder.
6. Unlike addition, subtraction, and multiplication, **division does not always produce a polynomial** — the remainder term $R(x)/Q(x)$ may persist.
7. If $R(x) = 0$, then $Q(x)$ divides $P(x)$ exactly, and $Q(x)$ is called a **factor** of $P(x)$.

Lecture 6 — Division Algorithm & Long Division Method

Terminology, the Formal Algorithm, and a Worked Example

1 Terminology

Naming the Parts of a Division

For $\dfrac{P(x)}{Q(x)} = S(x) + \dfrac{R(x)}{Q(x)}$:

Term	Name
$P(x)$ (numerator)	**Dividend**
$Q(x)$ (denominator)	**Divisor**
$S(x)$	**Quotient**
$R(x)$	**Remainder** (declared only when $\deg(R) < \deg(Q)$)

Rational function: Viewing polynomials as functions, $\dfrac{P(x)}{Q(x)}$ (with $Q(x) \neq 0$) is called a **rational function**.

2 The Formal Division Algorithm

Step-by-Step Procedure

1. **Arrange** both dividend and divisor in **descending order** of degree; insert missing powers with coefficient 0.
2. **Divide** the leading term of the current dividend by the leading term of the divisor → this gives the next term of the **quotient**.
3. **Multiply** this quotient term by the **entire divisor**.
4. **Subtract** the result from the current dividend → this gives the **new dividend**.
5. **Check**: is deg(new dividend) < deg(divisor)?
   - **Yes** → stop. The new dividend is the **remainder**.
   - **No** → repeat from Step 2 using the new dividend.

3 Worked Example: Long Division

Divide $2x^3 + 3x^2 + 1$ by $2x + 1$

Setup: Write the dividend with all powers present: $2x^3 + 3x^2 + 0x + 1$, divided by $2x + 1$. We show each round explicitly below.

Round 1

Divide leading terms: $\dfrac{2x^3}{2x} = x^2$. Multiply back: $x^2(2x+1) = 2x^3 + x^2$.

Subtract: $(2x^3 + 3x^2 + 0x + 1) - (2x^3 + x^2) = 2x^2 + 0x + 1$.

Check: $\deg(2x^2 + 0x + 1) = 2 \geq \deg(2x+1) = 1$. Continue.

Round 2

Divide leading terms: $\dfrac{2x^2}{2x} = x$. Multiply back: $x(2x+1) = 2x^2 + x$.

Subtract: $(2x^2 + 0x + 1) - (2x^2 + x) = -x + 1$.

Check: $\deg(-x+1) = 1 \geq \deg(2x+1) = 1$. Continue.

Round 3

Divide leading terms: $\dfrac{-x}{2x} = -\dfrac{1}{2}$. Multiply back: $-\dfrac{1}{2}(2x+1) = -x - \dfrac{1}{2}$.

Subtract: $(-x+1) - \left(-x - \dfrac{1}{2}\right) = 1 + \dfrac{1}{2} = \dfrac{3}{2}$.

Check: $\deg\left(\dfrac{3}{2}\right) = 0 < \deg(2x+1) = 1$. **Stop.**

Final Answer

Quotient: $S(x) = x^2 + x - \dfrac{1}{2}$. **Remainder**: $R(x) = \dfrac{3}{2}$.

$$\boxed{\frac{2x^3 + 3x^2 + 1}{2x+1} = x^2 + x - \frac{1}{2} + \frac{3/2}{2x+1}}$$

Verification: $\left(x^2 + x - \dfrac{1}{2}\right)(2x+1) + \dfrac{3}{2} = 2x^3 + 3x^2 + 0x + 1$. ✓

4 Summary Table of the Rounds

Round	Quotient term	Multiply back	New dividend
1	x^2	$2x^3 + x^2$	$2x^2 + 0x + 1$
2	x	$2x^2 + x$	$-x + 1$
3	$-\frac{1}{2}$	$-x - \frac{1}{2}$	$\frac{3}{2}$ (remainder)

Common pitfall: When subtracting a term like $-\dfrac{1}{2}(2x+1) = -x - \dfrac{1}{2}$, watch the sign carefully — subtracting a negative becomes addition. This is the step where sign errors most often creep in during long division.

5 Key Takeaways — Lecture 6

1. **Terminology**: dividend $P(x)$, divisor $Q(x)$, quotient $S(x)$, remainder $R(x)$. The remainder is declared only once $\deg(R) < \deg(Q)$.
2. $P(x)/Q(x)$ is called a **rational function** when viewed as functions, with $Q(x) \neq 0$.
3. **Division algorithm**: arrange in descending order (padding zeros), then repeatedly divide-multiply-subtract on leading terms until the remainder's degree drops below the divisor's degree.
4. **Worked example**: $(2x^3 + 3x^2 + 1) \div (2x + 1) = x^2 + x - \frac{1}{2}$ with remainder $\frac{3}{2}$.
5. **Sign care**: multiplying the quotient term by the divisor and subtracting is where errors most commonly occur — double-check signs at each round.
6. This long division method is exactly equivalent to the earlier "divide, multiply, subtract" strategy — long division just organizes the bookkeeping more clearly.

Lecture 7 — Recognizing Polynomial Graphs

Smoothness and Continuity as Visual Tests

1 The Twofold Mission

Two Complementary Questions

1. **Given a graph**, can we tell if it represents a polynomial function?
2. **Given a polynomial equation**, can we sketch its graph?

This lecture tackles Question 1; later lectures build tools for Question 2.

2 Two Visual Tests

Recall linear and quadratic graphs (lines and parabolas) — both can be drawn **without lifting the pen** and **without any sudden jerks**. These two observations generalize to **all** polynomial functions.

Test 1: Smoothness (No Sharp Corners)

A polynomial graph is always a **smooth curve** — it has no sharp corners, kinks, or cusps. At every point, the curve turns **gradually**, never abruptly.
If a graph has even one sharp corner, it **cannot** be a polynomial function.

Test 2: Continuity (No Breaks)

A polynomial graph can always be traced **without lifting the pen** — there are no breaks, jumps, or gaps.
If a graph has any break (the curve stops and restarts elsewhere), it **cannot** be a polynomial function.

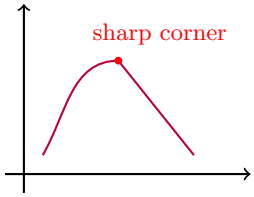
Combined Rule (Imp)

$$\boxed{\text{Polynomial graph} \implies \text{Smooth } \textbf{and} \text{ Continuous}}$$

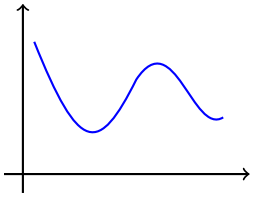
$$\boxed{\text{Smooth and Continuous} \not\Longrightarrow \text{Polynomial Function Always}}$$

Both conditions must hold. Failing **either** disqualifies the graph from being a polynomial.

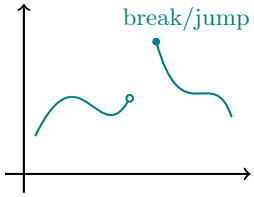
3 Visual Examples



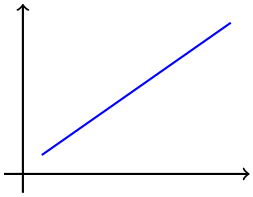
Graph A: Not smooth (sharp corner) ⇒ **NOT** a polynomial.



Graph B: Smooth and continuous ⇒ **IS** a polynomial.



Graph C: Smooth pieces but **discontinuous** (jump) ⇒ **NOT** a polynomial.



Graph D: A line — smooth and continuous ⇒ **IS** a polynomial (linear).

4 Applying the Tests: Four Graphs

Graph	Description	Polynomial?	Reason
1	Smooth, wavy, no breaks	**Yes**	Passes both tests: smooth and continuous
2	Smooth mostly, one sharp corner	**No**	Fails smoothness test at the corner point
3	A straight line	**Yes**	Smooth and continuous (linear polynomial)
4	Smooth pieces, but a jump/gap	**No**	Fails continuity test — pen must be lifted

5 Why These Tests Work

Polynomials are built from sums of terms like $a_k x^k$, each of which is smooth and continuous everywhere on $\mathbb{R}$ (no denominators, no roots, no absolute values). Since sums of smooth, continuous functions remain smooth and continuous, **every** polynomial function is smooth and continuous on its entire domain ($\mathbb{R}$).
Conversely, graphs with corners (like $|x|$, which is **not** a polynomial) or jumps (like step functions) signal the presence of non-polynomial behavior.

6 Practical Heuristic

Quick Recognition Guide

- Polynomial graphs often show several **ups and downs** (depending on degree) — this is a typical, though not required, feature.
- **No corners**: the curve should look "free-hand smooth" at every point.
- **No breaks**: you should be able to trace a polynomial function in a single stroke, pen never leaving the paper.

7 Looking Ahead

Having learned to **recognize** polynomial graphs visually, the next natural question is about **zeros** of a polynomial function — the points where the graph crosses the x-axis. This will be the focus of the next lecture.

8 Key Takeaways — Lecture 7

1. The mission is twofold: (a) identify polynomials from graphs, (b) sketch graphs from equations. This lecture addresses (a).
2. **Test 1 — Smoothness**: no sharp corners or kinks; the curve turns gradually everywhere.
3. **Test 2 — Continuity**: no breaks or jumps; drawable without lifting the pen.
4. A polynomial function must pass **both** tests.
5. **Why it works**: polynomials are sums of smooth, continuous power terms $a_k x^k$; sums of such terms remain smooth and continuous everywhere. .
6. **Next**: finding the **zeros** of a polynomial function (where the graph crosses the x-axis).

Lecture 8 — Zeros of Polynomial Functions

Finding x-Intercepts by Factoring

1 What Is a Zero of a Function?

Definition

If f is a polynomial function, a **zero of** f is any value of x for which $f(x) = 0$.
Graphically, zeros are exactly the **x-intercepts** — the points where the graph crosses (or touches) the x-axis.

2 Recap: Techniques from Quadratic Functions

For quadratics, we had two main approaches:

1. **Graphing**: plot points, use the axis of symmetry, read off where the curve meets the x-axis.
2. **Factoring**: write the quadratic in factored (intercept) form; each factor set to zero gives a root.

For general polynomials, graphing is often **impractical by hand**, so **factoring** becomes the central tool.

3 Common Factoring Techniques

Toolbox for Factoring Polynomials

Technique	Idea
Greatest Common Factor (GCF)	Pull out the largest monomial common to every term first.
Factor by grouping	Split terms into pairs/groups, factor each group, then factor out the common binomial.
Trinomial factoring	Treat a suitable block of terms as a quadratic-like trinomial (e.g., substitute $t = x^2$) and factor as usual.

4 Algorithm: Finding x-Intercepts by Factoring

Step-by-Step Strategy

1. Set $f(x) = 0$.
2. Look for a **greatest common monomial factor**; pull it out if present.
3. Factor the remaining expression into **binomials/trinomials** (grouping, quadratic-type substitution, etc.).
4. **Set each factor equal to zero** and solve — these solutions are the x-intercepts.

5 Worked Example 1

$f(x) = x^6 - 8x^4 + 16x^2$

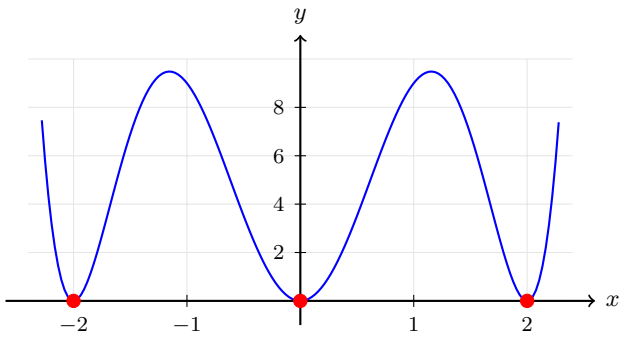
Step 1: $x^6 - 8x^4 + 16x^2 = 0$.
Step 2 (GCF): Common factor x^2:

$$x^2(x^4 - 8x^2 + 16) = 0$$

Step 3 (Substitution): Let $t = x^2$. Then $x^4 - 8x^2 + 16 = t^2 - 8t + 16 = (t-4)^2$.
Substituting back: $(x^2-4)^2 = 0 \Rightarrow x^2 - 4 = 0 \Rightarrow (x-2)(x+2) = 0$.
Full factorization:

$$x^2(x-2)^2(x+2)^2 = 0$$

Step 4: $x = 0$, $x = 2$, $x = -2$ (each a **repeated** root, multiplicity 2).
Step 5: Verified graphically — the curve touches the x-axis at $x = -2, 0, 2$ (touches rather than crosses, since each root has even multiplicity).



$f(x) = x^6 - 8x^4 + 16x^2$: touches (bounces off) the x-axis at $x = -2$, 0, 2

6 Worked Example 2: Factor by Grouping

$f(x) = x^3 - 4x^2 - 3x + 12$

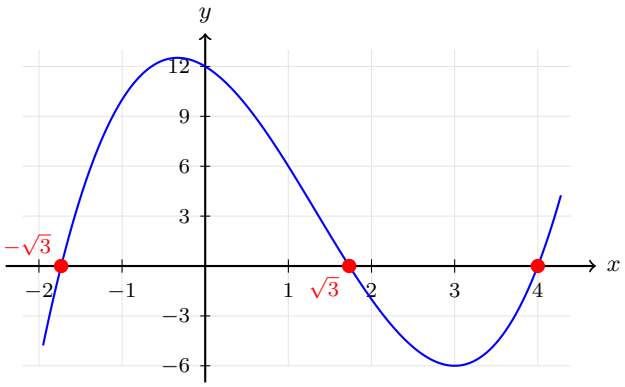
Step 1: $x^3 - 4x^2 - 3x + 12 = 0$.
Step 2 (GCF): No common monomial (constant term present).
Step 3 (Grouping): Group in pairs:

$$\underbrace{x^3 - 4x^2}_{x^2(x-4)} + \underbrace{-3x + 12}_{-3(x-4)} = x^2(x-4) - 3(x-4) = (x^2-3)(x-4)$$

Step 4: Set each factor to zero:

$$x^2 - 3 = 0 \Rightarrow x = \pm\sqrt{3}, \qquad x - 4 = 0 \Rightarrow x = 4$$

x-intercepts: $x = 4,\ \sqrt{3},\ -\sqrt{3}$.
Step 5: Verified graphically — three distinct crossings of the x-axis at $-\sqrt{3} \approx -1.73$, $\sqrt{3} \approx 1.73$, and 4.



$f(x) = x^3 - 4x^2 - 3x + 12$: crosses the x-axis at $x = -\sqrt{3},\ \sqrt{3},\ 4$

7 Worked Example 3: Already-Factored Form (Finding Both Intercepts)

$g(x) = (x-1)^2(x+3)$

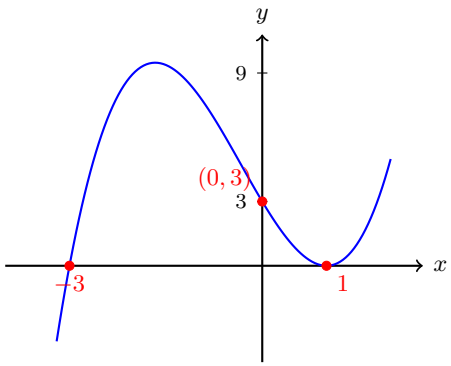
Since $g(x)$ is already factored, the x**-intercepts** are immediate:

$$g(x) = 0 \ \Rightarrow\ (x-1)^2 = 0 \text{ or } (x+3) = 0 \ \Rightarrow\ x = 1 \text{ (mult. 2)},\ x = -3$$

y**-intercept**: Set $x = 0$:

$$g(0) = (0-1)^2(0+3) = (1)(3) = 3$$

Summary: x-intercepts at $x = 1$ and $x = -3$; y-intercept at $y = 3$.



$g(x) = (x-1)^2(x+3)$: touches at x=1, crosses at x= − 3

8 Multiplicity: Touch vs. Cross

A Preview: Even vs. Odd Multiplicity

- If a root has **even multiplicity** (e.g., $(x-2)^2$), the graph **touches** the x-axis and turns back (like Example 1's roots).
- If a root has **odd multiplicity** (e.g., a simple factor $(x-4)$), the graph **crosses** the x-axis (like Example 2's roots).

This connects the algebra of factoring directly to the shape of the graph — a theme to be explored further in upcoming lectures.

9 Key Takeaways — Lecture 8

1. A **zero** of f is a value of x with $f(x) = 0$; graphically, these are the **x-intercepts**.
2. **Toolbox**: GCF extraction, factoring by grouping, and quadratic-type substitution (e.g., $t = x^2$) for trinomial-like expressions.
3. **Algorithm**: set $f(x) = 0 \to$ extract GCF $\to$ factor remaining expression $\to$ set each factor to zero $\to$ verify graphically.
4. Worked examples confirmed: $x^6 - 8x^4 + 16x^2 = 0$ gives roots $0, \pm 2$ (all multiplicity 2); $x^3 - 4x^2 - 3x + 12 = 0$ gives roots $4, \pm\sqrt{3}$; $(x-1)^2(x+3)$ gives x-intercepts $1, -3$ and y-intercept 3.
5. **Multiplicity preview**: even multiplicity $\to$ graph touches the axis; odd multiplicity $\to$ graph crosses the axis.

Lecture 9 — Multiplicity of Zeros: Touch or Cross?

Connecting Factor Powers to Graph Behavior

1 Where We Stand

So far we can: (1) recognize a polynomial from its graph, (2) find its zeros algebraically by factoring. The missing piece: knowing the zeros and intercepts **alone** doesn't tell us the full **shape** of the curve — where it rises, falls, or turns. This lecture adds a key tool: **multiplicity**.

2 Sketching by Hand: A Cubic Without Obvious Factors

Strategy When Factoring Isn't Obvious

If a cubic $f(x)$ has no visible factor or grouping pattern, use a **table of values** to hunt for zeros:

x	-2	-1	0	1	2
$f(x)$	0	$*$	$*$	0	$*$

Two zeros found by inspection: $x = -2$ and $x = 1$.

Finding the Third Root by Long Division

Since $x = -2$ and $x = 1$ are roots, $(x+2)$ and $(x-1)$ are factors. Multiply them to get a quadratic divisor:

$$(x+2)(x-1) = x^2 + x - 2$$

Long divide $f(x)$ by x^2+x-2 (from Lecture 6's method). The quotient reveals the last linear factor: $(x+3)$.
Full factorization (example): $f(x) = (x+3)(x+2)(x-1) = x^3 + 4x^2 + x - 6$.
A cubic has **at most 3** real roots — once we have 3 linear factors, factoring is complete.

Verification: $(x+3)(x+2)(x-1)$ expands to x^3+4x^2+x-6; substituting $x = -3, -2, 1$ each gives 0. ✓

3 Sketching with Three Known Roots

Plot the three x-intercepts $(-3, 0)$, $(-2, 0)$, $(1, 0)$ plus any table points, then connect smoothly. Near each simple root the curve **crosses** the axis — but without more theory, we can't yet predict exactly how many times the curve rises and falls between roots. That's what **multiplicity** explains.

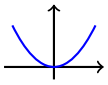
4 Multiplicity: How Many Times Does a Factor Repeat?

Definition

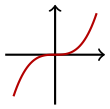
If a factor $(x-a)$ appears k times in the factorization of $f(x)$ — i.e., $f(x) = (x-a)^k \cdot$ (other factors) — then a is a zero of **multiplicity** k.

5 Case Study: $f(x) = (x-1)^2(x+2)^3(x+4)$

Factor	Zero	Multiplicity	Local shape
$(x-1)^2$	$x=1$	2 (even)	Like $y=x^2$ near this point — **bounces off**
$(x+2)^3$	$x=-2$	3 (odd)	Like $y=x^3$ near this point — **crosses**, flattens
$(x+4)^1$	$x=-4$	1 (odd)	Like $y=x$ near this point — **crosses**, straight-through



Even mult. (e.g., $(x-a)^2$): **touches**, bounces off



Odd mult. (e.g., $(x-a)^3$): **crosses**, flattens & decays

6 The General Rule

Even vs. Odd Multiplicity

Even multiplicity $\Rightarrow$ graph **touches**/is tangent to the x-axis (bounces off)

Odd multiplicity $\Rightarrow$ graph **crosses** the x-axis

Why: near $x=a$, the factor $(x-a)^k$ dominates the local behavior, mimicking $y=x^k$. Even powers of x (like x^2, x^4) stay non-negative near 0 — the curve cannot go below the axis, so it bounces. Odd powers (like x, x^3) change sign at 0, so the curve passes through.

7 Higher Multiplicities: Flatness Increases

Effect of Increasing Multiplicity

Multiplicity sequence	Visual effect near the zero
Even: $2, 4, 6, \ldots$	The "bounce" becomes **flatter and broader** at the base as multiplicity increases; still touches, never crosses.
Odd: $1, 3, 5, 7, \ldots$	The curve **flattens more** as it approaches and leaves the zero (looks momentarily horizontal), but still crosses.

Rule of thumb: the higher the multiplicity, the **flatter** the graph becomes at the roots — regardless of even/odd, more repetition means more flattening.

8 Putting It All Together: The Three-Skill Combo

Full Strategy for Understanding a Polynomial's Graph

1. **Recognize**: confirm the given curve is smooth and continuous (Lecture 7).
2. **Find zeros**: factor the polynomial (GCF, grouping, substitution, or long division) to locate all x-intercepts (Lecture 8).
3. **Determine multiplicity**: examine the power of each factor to predict **touch** (even) vs. **cross** (odd), and how flat the curve is near each zero (this lecture).

Together, these three skills let us sketch a very good approximation of a polynomial's graph **by hand**, without needing calculus (yet).

9 Key Takeaways — Lecture 9

1. When factoring isn't obvious, use a **table of values** to spot easy zeros, then **long divide** to find remaining factors.
2. **Multiplicity**: the number of times a factor $(x - a)$ repeats in the factorization.
3. **Even multiplicity** $\rightarrow$ graph **touches** the x-axis and bounces back (like $y = x^2$ locally).
4. **Odd multiplicity** $\rightarrow$ graph **crosses** the x-axis (like $y = x$ or $y = x^3$ locally).
5. **Higher multiplicity** (regardless of parity) $\rightarrow$ the curve **flattens more** near that zero.

Lecture 10 — Reverse Engineering: Graph to Multiplicities

Using the Degree Constraint to Pin Down Unknown Multiplicities

1 Formal Terminology

Multiplicity, Precisely

If $f(x)$ has a factor $(x-a)^m$, then m is called the **multiplicity** of the zero $x=a$.
Behavior at $x=a$:

- m **even** $\Rightarrow$ graph **touches** the x-axis (does not cross).
- m **odd** $\Rightarrow$ graph **crosses** the x-axis.

2 The Degree Constraint

Sum of Multiplicities $\leq$ Degree

If $f(x)$ has degree n with real zeros of multiplicities $m_1, m_2, \ldots, m_k$:

$$\boxed{m_1 + m_2 + \cdots + m_k \ \leq \ n}$$

Why not always equal? Not every root needs to be **real** — some roots may be **complex** (not visible on the real graph).

3 Worked Example 1: Degree 6, Three Visible Zeros

Reading the Graph

A degree-6 polynomial has visible x-intercepts at $x=-2, 0, 2$. Observed shapes:

Zero	Observed shape	Possible multiplicity
$x=-2$	Straight-line crossing	1 (odd, simplest case)
$x=0$	Twisted "S" shape	Odd: $3, 5, 7, \ldots$
$x=2$	Bounces off	Even: $2, 4, 6, \ldots$

Using the degree constraint: multiplicities must sum to **exactly** 6 (since we can account for all intercepts visibly, assuming no hidden complex roots):

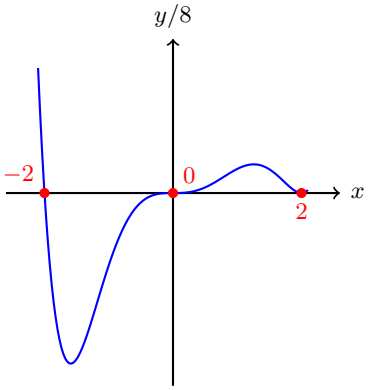
$$1 + m_0 + m_2 = 6 \quad \Rightarrow \quad m_0 + m_2 = 5$$

Try $m_0 = 5$: then $m_2 = 0$ — impossible (need $m_2 \geq 2$, even). **Rejected.**
Try $m_0 = 3$: then $m_2 = 2$. ✓Consistent (odd, even as required).

Result

$$x = -2: \text{ mult. } 1, \qquad x = 0: \text{ mult. } 3, \qquad x = 2: \text{ mult. } 2$$

Proposed form: $f(x) = a \cdot x^3(x-2)^2(x+2)$, a degree-6 polynomial.
Verification: $x^3(x-2)^2(x+2)$ expands to $x^6 - 2x^5 - 4x^4 + 8x^3$ — degree 6. ✓



$f(x) = x^3(x-2)^2(x+2)$ (y-axis compressed $\times\frac{1}{8}$ to fit)

The Unknown Scaling Constant a

Knowing the **factors** does not pin down the exact polynomial — a scaling constant a remains unknown:

$$f(x) = a \cdot x^3(x-2)^2(x+2)$$

The factors and their multiplicities are determined by the x-intercepts and touch/cross behavior, but a requires an **additional point** on the graph (e.g., a y-intercept value) to solve for exactly. This will be addressed in a later lecture.

4 Worked Example 2: Degree 4, Only One Visible Zero

A Trickier Case

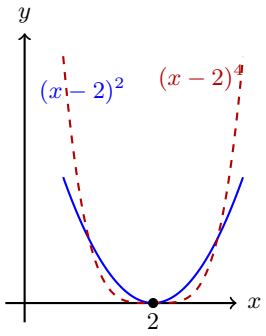
A degree-4 polynomial shows only **one** visible x-intercept, at $x = 2$, where the graph **bounces off** (even multiplicity).
Possible multiplicities: 2 or 4 (must be even, at most equal to the degree).

Ruling Out Multiplicity 4

Compare the actual graph's shape near $x = 2$ with the reference shapes:

- $(x-2)^2$: standard parabola-like bounce.
- $(x-2)^4$: a **much flatter**, broader bounce (higher multiplicity flattens the curve, as seen in Lecture 9).

The given graph's bounce matches $(x-2)^2$ much more closely — **not** the broader, flatter shape of $(x-2)^4$. So we conclude multiplicity $= 2$.



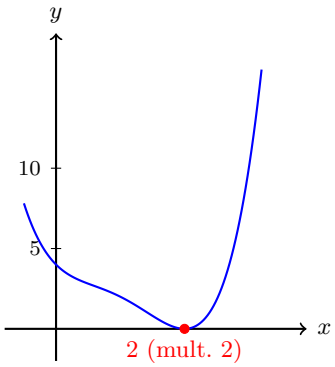
The actual graph's bounce matches the narrower $(x-2)^2$ shape

Where Did the Other Two Roots Go?

Since the polynomial has degree 4 but only 2 real roots are accounted for ($m = 2$ at $x = 2$), the remaining **2 degrees are "missing."**
Explanation: The complete factorization is

$$f(x) = (x^2 + 1)(x - 2)^2$$

The factor $x^2 + 1$ has **no real roots** (its roots are $x = \pm i$, complex). These roots exist in the complex domain and are simply **invisible** on the real xy-graph.
Verification: $(x^2 + 1)(x - 2)^2 = x^4 - 4x^3 + 5x^2 - 4x + 4$, degree 4, with real solution set $\{2\}$ (multiplicity 2) and complex solutions $\pm i$.



$f(x) = (x^2+1)(x-2)^2$: only real zero is x=2; the x^2+1 factor stays positive

5 Resolving the "$\leq$" in the Degree Constraint

Why the Inequality, Not Equality

For k **distinct real** zeros with multiplicities $m_1, \ldots, m_k$ of a degree-n polynomial:

$$m_1 + m_2 + \cdots + m_k \leq n$$

Equality holds when **all** roots are real (every factor is linear over $\mathbb{R}$).
Strict inequality holds when some roots are **complex** — these contribute to the degree but not to any visible x-intercept.
In Example 2: $2 \leq 4$ (strict), because 2 degrees are "used up" by the complex pair $\pm i$.

6 Summary: Reading a Graph

Checklist for Reverse-Engineering a Polynomial Graph

1. Identify every visible x-intercept.
2. At each intercept, classify the local behavior: **touch** (even mult.) or **cross** (odd mult.).
3. Use the **shape sharpness/flatness** to narrow down the exact even or odd multiplicity (compare to x^2 vs. x^4, or x vs. x^3, etc.).
4. Check that the **sum of multiplicities does not exceed** the given degree n.
5. If the sum is **less than** n, the remaining degree corresponds to **complex (non-real) roots**, invisible on the graph.
6. The scaling constant a in $f(x) = a \cdot (\text{factors})$ cannot be determined from shape alone — an extra point (e.g., y-intercept) is needed.

7 Key Takeaways — Lecture 10

1. **Multiplicity** m: exponent on a factor $(x-a)^m$. Even $\rightarrow$ touch; odd $\rightarrow$ cross.
2. **Degree constraint**: $\sum m_i \leq n$, with equality only when all roots are real.
3. **Worked Example 1**: degree-6 graph with 3 visible zeros; degree constraint plus shape observations uniquely pin down multiplicities $1, 3, 2$.
4. **Worked Example 2**: degree-4 graph with only 1 visible zero (mult. 2); the missing 2 degrees belong to a complex-conjugate pair ($x^2 + 1$ contributes $\pm i$).
5. **Shape comparison** (flatness) helps distinguish between candidate even (or odd) multiplicities, e.g., $(x-2)^2$ vs. $(x-2)^4$.
6. Knowing the factored form up to multiplicities still leaves an **unknown scale factor** a; an additional point on the graph is needed to fully determine $f(x)$.
7. Complex roots always occur, and contribute to the degree, but never appear as x-intercepts on a real graph.

Lecture 11 — End Behavior of Polynomials

What Happens Far Away from All the Roots?

1 The Missing Piece

We can now find zeros, multiplicities, and local shape near each root. But one question remains: **after passing through all the roots**, does the graph shoot up to $+\infty$, or down to $-\infty$, on each side? This is called the **end behavior** of the polynomial.

2 Recap: Quadratics

For $f(x) = a_2x^2 + a_1x + a_0$:

- $a_2 > 0$: parabola opens **upward** — both ends go to $+\infty$.
- $a_2 < 0$: parabola opens **downward** — both ends go to $-\infty$.

The end behavior of a quadratic is decided entirely by its **highest-degree term**. This generalizes to all polynomials.

3 Why the Leading Term Dominates

The Key Insight

For $f(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0$ (with $a_n \neq 0$):
As $|x|$ becomes very large (whether $x \to +\infty$ or $x \to -\infty$), the term a_nx^n grows **much faster** than any lower-degree term. It **dominates** the sum, so the overall sign and growth of $f(x)$ matches that of a_nx^n alone.

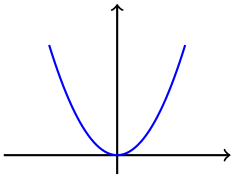
$$\boxed{\text{End behavior of } f(x) \text{ is determined entirely by } a_nx^n}$$

4 The Leading Coefficient Test

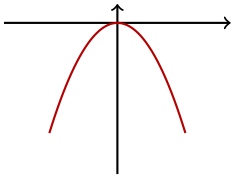
Four Cases Based on Parity of n and Sign of a_n

Degree n	**Sign of a_n**	**As $x \to -\infty$**	**As $x \to +\infty$**	**Model**
Even	$a_n > 0$	$f(x) \to +\infty$	$f(x) \to +\infty$	like x^2
Even	$a_n < 0$	$f(x) \to -\infty$	$f(x) \to -\infty$	like $-x^2$
Odd	$a_n > 0$	$f(x) \to -\infty$	$f(x) \to +\infty$	like x^3
Odd	$a_n < 0$	$f(x) \to +\infty$	$f(x) \to -\infty$	like $-x^3$

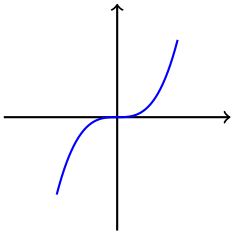
5 Visualizing the Four Cases



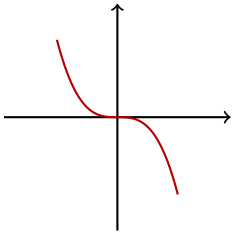
Even, $a_n > 0$: both ends up $\uparrow\uparrow$ (like x^2).



Even, $a_n < 0$: both ends down $\downarrow\downarrow$ (like $-x^2$).



Odd, $a_n > 0$: down-left, up-right $\downarrow\uparrow$ (like x^3).



Odd, $a_n < 0$: up-left, down-right $\uparrow\downarrow$ (like $-x^3$).

6 Connecting End Behavior to Odd-Degree Crossings

Recall: polynomials of **odd degree** always **cross** the x-axis (Lecture 9). This connects naturally to end behavior: an odd-degree polynomial **must** go from $-\infty$ to $+\infty$ (or vice versa) as x sweeps across $\mathbb{R}$, so it is forced to cross zero at least once — consistent with the crossing behavior at odd-multiplicity roots.
Even-degree polynomials, by contrast, can have **both ends pointing the same direction** (both up or both down), consistent with the possibility of **no real roots at all** (e.g., $x^2 + 1$ from Lecture 10).

7 Key Takeaways — Lecture 11

1. **End behavior**: what $f(x)$ does as $x \to \pm\infty$, after all roots have been passed.
2. The **leading term** $a_n x^n$ alone determines end behavior — it dominates all lower-degree terms for large $|x|$.
3. **Odd-degree** polynomials always have ends pointing in **opposite** directions, consistent with them always crossing the x-axis at least once.
4. **Even-degree** polynomials have ends pointing the **same** direction, consistent with the possibility of having **no** real roots.
5. From a graph alone, end behavior reveals the **parity of the degree** and the **sign of the leading coefficient** — key information for reconstructing the polynomial.

Lecture 12 — Turning Points of Polynomials

Counting the Twists Between Zeros

1 What Is a Turning Point?

Definition

A **turning point** is a point on the graph where the function changes behavior:

- from **increasing to decreasing** (rising to falling), or
- from **decreasing to increasing** (falling to rising).

Note: A precise, rigorous identification of turning points requires **calculus** (We will look later in this course). For now, we use a rough, algebra-based estimate.

2 Connecting Degree to Turning Points

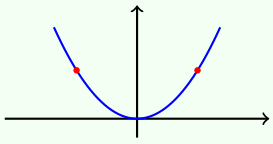
The Reasoning

A degree-n polynomial has **at most** n **real zeros**. To pass through n distinct zeros, the curve must weave up and down — every time it returns to cross the axis again, it must have turned around at least once in between.

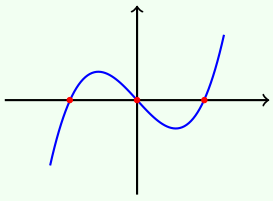
Quadratic (2 zeros, if real): needs 1 turn (the vertex).

Cubic (up to 3 zeros): needs up to 2 turns.

Quartic (up to 4 zeros): needs up to 3 turns.



Quadratic: 2 zeros, 1 turn



Cubic: 3 zeros, 2 turns

3 The General Hypothesis

Maximum Turning Points

For a polynomial of degree n:

$$\boxed{\text{Maximum number of turning points} = n - 1}$$

This is a **hypothesis** at this stage (an upper bound based on counting zeros); a rigorous proof requires calculus. It is, however, extremely useful as a **sanity check** when sketching graphs.

4 Using Turning Points as a Sanity Check

Example 1: $f(x) = 1 + x^2 + 4x^5$

Degree $n = 5 \Rightarrow$ at most $n - 1 = 4$ turning points.
The actual graph shows **only 2** turning points. Since $2 \leq 4$, the graph is **consistent** with the hypothesis. (It need not achieve the maximum — the bound only says "at most.")
Warning: If a sketch of this function showed **5 or more** turning points, we would immediately know the sketch is **wrong**.

Example 2: $f(x) = (x-1)^3(x+2)$

Degree $n = 4$ (since $(x-1)^3$ contributes 3 and $(x+2)$ contributes 1) $\Rightarrow$ at most $n - 1 = 3$ turning points.
Zeros: $x = 1$ (multiplicity 3, odd $\rightarrow$ crosses) and $x = -2$ (multiplicity 1, odd $\rightarrow$ crosses).
The graph shows only **one clear turning point**, located between the two zeros. But at $x = 1$, something unusual happens — the curve flattens dramatically (like a "shelf") before continuing to rise. This is **not** classified as a turning point under our current definition, because the function does **not** switch from increasing to decreasing there — it keeps increasing through that point, just very slowly.

5 Distinct vs. Repeated Roots: Verifying the Hypothesis

Example 3: $f(x) = (x-2)(x-4)(x+3)$ — **All Distinct Roots**

Degree $n = 3$, three **distinct** zeros: $x = 2, 4, -3$.
To pass through all three distinct zeros, the graph **must** turn around exactly $n - 1 = 2$ times — and indeed it does. This example **achieves** the maximum, confirming the hypothesis holds with equality when all roots are real and distinct.

6 Why the Repeated-Root Case Is Subtle

The Special Point Revisited

For $f(x) = (x-1)^3(x+2)$, calculus reveals:

$$f'(x) = (x-1)^2(4x+5)$$

The factor $(x-1)^2$ is **never negative**, so it does **not** cause $f'(x)$ to change sign at $x = 1$ — the slope touches 0 there but the function keeps increasing on both sides. Only at $x = -\frac{5}{4}$ (from the factor $4x + 5$) does the slope actually change sign, giving the **one true turning point**.
Preview: In calculus, points where $f'(x) = 0$ are called **critical points**. Not every critical point is a turning point — some (like $x = 1$ here) are **stationary inflection points**, where the graph flattens momentarily but does not reverse direction.

7 Summary of the Three Examples

Function	**Degree n**	**Max $(n-1)$**	**Actual**	**Note**
$1 + x^2 + 4x^5$	5	4	2	Fewer than max — still valid
$(x-1)^3(x+2)$	4	3	1	Repeated root hides a "flat" point
$(x-2)(x-4)(x+3)$	3	2	2	All distinct roots — achieves max

8 Key Takeaways — Lecture 12

1. A **turning point** is where the graph switches between increasing and decreasing.
2. **Hypothesis**: a degree-n polynomial has **at most** $n-1$ turning points. Rigorous proof will be given in later weeks, but the bound is a valuable **sanity check** for hand-drawn graphs.
3. Actual turning points can be **fewer** than $n-1$ (e.g., $1 + x^2 + 4x^5$ has only 2, though up to 4 are allowed).
4. When all real roots are **distinct**, the polynomial tends to **achieve** the maximum $n-1$ turning points (e.g., $(x-2)(x-4)(x+3)$).
5. **Repeated roots** can hide subtlety: at $x = 1$ in $(x-1)^3(x+2)$, the slope is momentarily zero but doesn't change sign — so it is **not** counted as a turning point by this definition, even though something visually notable happens there.

Lecture 13 — Putting It All Together: Graphing & Reverse-Engineering

The Complete Algorithm, the Intermediate Value Theorem, and Finding $f(x)$ from a Graph

1 The Complete Graphing Algorithm

Step-by-Step Recipe (Equation → Graph)

1. **Find intercepts**: x-intercepts (may not all be real) and y-intercept ($f(0)$).
2. **Check symmetry**: $f(-x) = f(x)$ (even, symmetric about y-axis) or $f(-x) = -f(x)$ (odd, symmetric about origin).
3. **Find zeros and multiplicities**: determines touch (even) vs. cross (odd) behavior at each intercept.
4. **Determine end behavior**: from the leading term $a_n x^n$ (Lecture 11).
5. **Estimate turning points**: at most $n - 1$ (Lecture 12); exact locations need calculus.
6. **Sketch**, combining all the above; then **verify with technology** (e.g., Desmos).

2 Symmetry: A Useful Shortcut

Even and Odd Symmetry

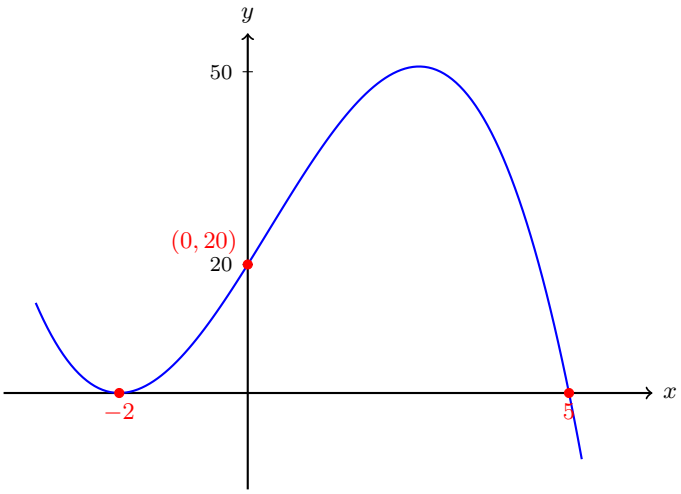
Type	Condition	Symmetry
Even function	$f(-x) = f(x)$	symmetric about the y-axis
Odd function	$f(-x) = -f(x)$	Symmetric about the origin (180° rotation)

Examples: $y = x^2$ is even (symmetric about y-axis). $y = x^3$ is odd (symmetric about origin).
Benefit: once you know the graph for $x > 0$, symmetry gives you the graph for $x < 0$ **for free** — no extra computation needed.

3 Worked Example: Sketching $f(x) = -(x+2)^2(x-5)$

Step-by-Step Sketch

x-intercepts: $x = -2$ (multiplicity 2, even → touches) and $x = 5$ (multiplicity 1, odd → crosses).
Degree and leading coefficient: expanding gives degree 3 with leading term $-x^3$, so $a_3 < 0$ (odd degree, negative leading coefficient).
End behavior: as $x \to +\infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to +\infty$.
y-intercept: $f(0) = -(0+2)^2(0-5) = -(4)(-5) = 20$.
Turning points: cubic ⇒ at most $3 - 1 = 2$ turning points.



$f(x) = -(x+2)^2(x-5)$: touches at -2, crosses at 5, y-intercept 20

4 The Intermediate Value Theorem (IVT)

Statement

Let f be a polynomial function (hence **continuous** everywhere). If $f(a)$ and $f(b)$ have **opposite signs** (say $f(a) > 0$ and $f(b) < 0$, with $a < b$), then there exists at least one $c \in (a, b)$ such that:

$$\boxed{f(c) = 0}$$

Intuition: since the graph is continuous (no breaks), to go from a positive value to a negative value, it **must** cross zero somewhere in between.

Why IVT Is Useful

When factoring is difficult, evaluate f at a few points. If you find $f(a)$ and $f(b)$ with **opposite signs**, IVT **guarantees** a root lies between a and b — even if you can't pin down its exact value algebraically. This builds confidence and narrows the search for roots (e.g., via further tabulation or numerical methods).

5 Reverse Direction: From Graph to Formula

The Reverse-Engineering Algorithm

1. **Read off** the x-intercepts from the graph.
2. **Determine multiplicity** of each (touch $\rightarrow$ even; cross $\rightarrow$ odd; use flatness to narrow down further).
3. **Check end behavior** to determine overall parity of degree and sign of the leading term.
4. **Write the least-degree polynomial** consistent with all factors found: $f(x) = a \cdot$ (product of factors).
5. **Use one more point** (usually the y-intercept) to solve for the unknown **stretch factor** a.

6 Worked Example: Finding $f(x)$ from a Graph

Given: x-intercepts at $-2, 1$; y-intercept at -2; two turning points

Step 1–2: $x = -2$ appears to **bounce** (even multiplicity); $x = 1$ appears **linear/crossing** (odd multiplicity, simplest guess: 1).
Step 3: Two turning points $\Rightarrow$ least degree is $n - 1 = 2 \Rightarrow n = 3$ (cubic); the graph crosses at both ends, consistent with **odd** degree.
Step 4: Simplest consistent factorization using the two intercepts with the correct multiplicities (even at -2, odd at 1), totaling degree 3:

$$f(x) = a(x+2)^2(x-1)$$

Step 5: Use y-intercept $f(0) = -2$:

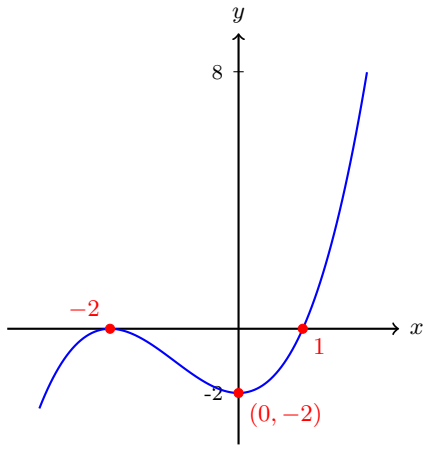
$$f(0) = a(0+2)^2(0-1) = a(4)(-1) = -4a$$

$$-4a = -2 \;\Rightarrow\; a = \frac{1}{2}$$

Final Answer

$$\boxed{f(x) = \frac{1}{2}(x+2)^2(x-1)}$$

Verification: Expanding gives $f(x) = \frac{1}{2}x^3 + \frac{3}{2}x^2 - 2$. Check: $f(0) = -2$ ✓, $f(-2) = 0$ ✓, $f(1) = 0$ ✓. Matches the graph when plotted.



Reconstructed: $f(x) = \frac{1}{2}(x+2)^2(x-1)$ — matches the original graph

7 The Complete Picture: Two Mirror-Image Skills

Direction	Process
Equation → Graph	Intercepts, symmetry, multiplicities, end behavior, turning points → sketch
Graph → Equation	Read intercepts and multiplicities from the picture, determine least degree, then solve for the stretch factor a using one extra point

These two skills are **inverses** of each other and together complete our study of graphing polynomial functions using pre-calculus tools.

8 Key Takeaways — Lecture 13

1. **Full graphing algorithm**: intercepts → symmetry → zeros/multiplicities → end behavior → turning points → sketch → verify.
2. **Symmetry shortcuts**: even functions ($f(-x) = f(x)$) mirror about the y-axis; odd functions ($f(-x) = -f(x)$) are symmetric about the origin.
3. Worked example confirmed: $f(x) = -(x+2)^2(x-5)$ has y-intercept 20 (watch signs carefully!), end behavior $+\infty \to -\infty$, and at most 2 turning points.
4. **Intermediate Value Theorem**: if $f(a)$ and $f(b)$ have opposite signs, a root exists strictly between a and b — a powerful existence tool when exact factoring fails.
5. **Reverse algorithm**: read intercepts/multiplicities from a graph, find least-degree factored form, then solve for the unknown stretch factor a using one additional point (usually the y-intercept).
6. Worked example confirmed: intercepts at -2 (even) and 1 (odd), y-intercept -2, two turning points $\Rightarrow f(x) = \frac{1}{2}(x+2)^2(x-1)$.

7. These two algorithms — equation-to-graph and graph-to-equation — are complementary and complete the toolkit for understanding polynomial functions without calculus.

References & Acknowledgements

These notes are based on the video lectures from the **Mathematics I** course offered at **IIT Madras BS Degree Programme**.

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